

Zhe Wei Kho

Nationality: Malaysia

I am a mechanical engineer with experience in advanced manufacturing. I bring a strong background in research, manufacturing, project management, and technical project leadership, with an understanding of supply chains, financial evaluation and lean principles. Analytical and curious, I excel in cross-disciplinary teams through a flexible approach and aim to continuously improve.

+44 7818 949924 | zhewei.kho@gmail.com | [linkedin.com/in/zhe-wei-kho](https://www.linkedin.com/in/zhe-wei-kho) | Project Portfolio: zhewei-k.github.io

EDUCATION

Imperial College London, United Kingdom

2025 – Present

MSc Composites: The Science, Technology and Engineering Application of Advanced Composites

Relevant Modules – Advanced Manufacturing, Lightweight Structures, Structural Integrity & Health Monitoring, view [full list](#)

- IMechE Postgraduate Masters Scholarship – £5,000
- MSc Design Challenge with Collins Aerospace – highest grade in group project (77.0%)
- Thesis Project: “High-Performance Thermoplastic Composite EM Shielding for Avionics” with Evonik Op. GmbH

The University of Sheffield, United Kingdom

2022 – 2025

BEng (Hons) Mechanical Engineering

First-Class (Top 5% in cohort)

Relevant Modules – Manufacturing Systems (87%), Advanced Mathematics (87%), Fluids Engineering (85%), Materials Processing (84%), Computational and Numerical Methods (80%), Advanced Thermodynamics (78%), view [full list](#)

- International Undergraduate Merit Scholarship – £37,500
- Faculty of Engineering Excellence Scholarship – £5,000
- Sheffield Engineering Leadership and Service Award (SELSA) – leadership academy
- Edgar Baidon Prize in Mechanical Engineering – highest grade in mechanical design modules (80.2%)

KEY SKILLS

CAD Design: Autodesk Fusion 360 CAD, nTop, SOLIDWORKS, AutoCAD, BS:8888 Drawings

Simulation and Analysis: Ansys Mechanical APDL FEA, Autodesk Fusion 360 FEA, Python, MATLAB, Simulink, PostgreSQL, MS Excel VBA Code, Git

Manufacturing: Additive (Polymer FDM, Metal SLS), Composites (Wet Layup, Prepreg with Autoclave, Chopped Fibre), Subtractive (Laser/Water Jet Cutting)

Lab System Integration: Arduino, Raspberry Pi

Laboratory: Mechanical Characterisation (Instron 50kN), Impact Damage Sensing (Composite panel)

Project Delivery: Leadership, Mentoring, Client Liaison and Engagement, Supply Chain, Pitching/Presentation

H&S and Documentation: FMEA, IOSH Risk Assessment, COSHH, MS Office (Word, Powerpoint, Excel), LaTeX

Languages: English (Native), Chinese (Professional), Malay (Professional), German (Beginner)

PROFESSIONAL ENGINEERING EXPERIENCE

DJI DronesKaki Sarawak

One of Malaysia's largest drone distributors, service and training providers for DJI. <https://droneskaki.com/>

Enterprise Operations Intern

Aug 2025 – Sep 2025

- Conducted **field deployment**, including large mapping operations at quarries, agriculture pest management, and state-of-the-art [government](#) building maintenance using DJI Enterprise M350, M400, and T50 drones.
- Implemented structured workflows for improved satisfaction and 60% reduction in response times; designed seven posters to simplify information for customers; showing **consideration for client needs**.

2H Offshore

Renewable energy marine engineering company serving clients like Shell, Wood, and Baker Hughes. <https://2hoffshore.com/>

Structural Engineer Intern

May 2022 – Aug 2022

- Designed spreadsheets to **analyse pad eyes and cathodic protection systems** with MS Excel VBA Macros to automate processes; worked with [PTS 11.22.02](#) and [DNVGL-RP-B401](#) standards.
- Verified feasibility and accuracy of engineering drawings for **offshore riser clamp structures** in SOLIDWORKS and Ansys Mechanical FEA for client delivery; worked with [ISO fasteners](#).
- Researched and presented a **coding framework for automating simulations**, reducing simulation times to less than 15 minutes; implemented using Ansys Mechanical APDL FEA in Python; further developed during summer research and undergraduate [thesis](#).

Kidocode

Malaysia's No.1 software engineering, business, and creativity school for all age groups. <https://www.kidocode.com/>

Additive Manufacturing Intern

Jan 2022 – Mar 2022

- Published an [article](#) on prototyping covering CNC, FDM, SLS which was **accepted into publication with 35k+ followers**; validated additive manufacturing course for deployment into in-house teaching platform.
- Taught over 200 students, teaching HTML, CSS, Python, and C++ code, amassing over 180 teaching hours.

ENGINEERING PROJECTS

Thermoplastic Composite Electromagnetic Shielding - Evonik Operations GmbH

Jan 2026 – Sep 2026

MSc Thesis Project – research, experiment design, product development. [report](#)

- Researched thermoplastic composite laminate enclosures to protect sensitive electronics from high-power electromagnetic waves, achieving 2x effectiveness compared to industry standard.
- Experimentally characterised physical, mechanical, electrical, and shielding properties in line with ASTM standards; developed modified ASTM D5568 test methodology to obtain EM shielding effectiveness.

Thermoplastic Composite Aircraft Seat Frame - Collins Aerospace UK

Jan 2026 – Mar 2026

Module Project – materials selection, finite element analysis, manufacturing analysis. [report](#)

- Developed a 16G economy class aircraft seat frame according to ETSO-C127 / 14 CFR 25.562 made from 90% composite materials, achieving a mass reduction of 51% (compared to standard aluminium frame).
- Conducted supply chain analysis on suppliers (Toray, Hexcel, Mitsubishi) and manufacturers (Pinette PEI, Airbus) to meet annual production rate of 36,750 units using CF/PEEK thermoplastic composites.
- Modelled ROI of 1.84 years for Airbus A320neo using British Airways London-Barcelona route.

Structural Health Monitoring

Oct 2025 – Jan 2026

Module Project – research, signal processing, . [report](#)

- Implemented modified RAPID algorithm in MATLAB for active sensing of simulated damage positions within Toray T300/914 carbon fibre composite laminates using piezoelectric transducers in diamond arrangement.
- Developed 11-page feasibility study, analysing data and proposing improvements to algorithm and setup.

Mechanical Testing of Composites

Oct 2025 – Jan 2026

Module Project – research, experiment design. [report](#)

- Researched CRAG, ASTM, and ISO standards and experimentally analysed unidirectional fibre composite materials; measuring tensile (CRAG 300), compression (ASTM D6641), shear (ASTM D5379), flexural (ASTM D790), and fracture toughness (ASTM D5528 and ISO 15114)

Lead Structural Design Engineer - Project Hex

Jul 2023 – Jul 2024

Student Project – project management, product development. [report](#)

- **Led a team of 10 students** to develop the structure of Vulcan V3, a 20kg, 3.2-meter wingspan eVTOL drone capable of deploying five medical packages; achieved overall mass reduction of 20%.
- Analysed carbon fibre fuselage under static load case in Ansys Mechanical FEA and vibrational analysis of carbon fibre beams supporting vertical liftoff motors, demonstrating
- Developed a winch-operated delivery system controlled via Arduino Nano and Raspberry Pi 4; achieved >40 drop tests from 15 meters within 6 months, assessed risk in-line with IOSH safety standards.
- Developed 10-page standard operating procedure (SOP) for manufacturing wet layup carbon fibre composites, including FDM additive manufactured moulds, reduced lead times by 83% down to 1 week.

LEADERSHIP EXPERIENCE

Malaysian and Singaporean Night (MNight)

Award-winning student-led theatrical production showcasing Malaysian culture. <https://instagram.com/sheffieldmnight>

Technical Director

Sep 2023 – April 2024

- Developed and executed 150+ lighting, backdrop, and audio cues simultaneously in collaboration with the Student Union Technical Team for a 3-hour theatrical production, demonstrating multitasking capabilities.
- Directed 70+ hours of rehearsals comprising 30+ scenes and 10+ musical performances, over 5 months.

Young Malaysian Engineers (YME)

Engineering student body connecting Malaysian students worldwide to industry. <https://www.ymeglobal.org/>

Vice President of UK Chapter

Oct 2022 – Jan 2024

- Led a team of 40+ in event planning, hosting a **2-week case study challenge with Intel**; attracting 120+ participants across Malaysia and the UK.
- Introduced data-driven decisions for recruitment, monthly town hall meetings, and establishing exclusive internal events with partners such as PwC; implemented a constant improvement attitude.
- Managed **global STEM outreach events**, such as the Malaysia-UK Students Outreach Fair (MUSOF) and the Malaysian Students' Technology Conference, with a total turnover of over 13,000 MYR.

VOLUNTEERING AND INTERESTS

Volunteering: SKY Career Empowerment Day (50 students), MCKL Engineering Society Workshop (20 students), Trash Hero (+130 kg of trash collected), MCKL Peer Support Group (delivered 14 flood victims).

Freelance Tutoring: Cambridge GCSE and A-Levels (helped one student from C to A* in 2 months).

Hobbies: Running (3x 21km finisher), Reading for self-development, Personal website and web-app development.